

BEE 4750 Homework 2: Systems Modeling and Simulation

Name:

ID:

Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `c:\Users\elrae\BEE4750\hw2-holly-1`

Installed JpegTurbo_jll	v3.1.3+0
Installed InlineStrings	v1.4.5
Installed NetworkLayout	v0.4.10
Installed Libmount_jll	v2.41.1+0
Installed OffsetArrays	v1.17.0
Installed Accessors	v0.1.42
Installed Preferences	v1.5.0
Installed EarCut_jll	v2.2.4+0
Installed Fontconfig_jll	v2.17.1+0
Installed ArnoldiMethod	v0.4.0
Installed StaticArrays	v1.9.15
Installed InvertedIndices	v1.3.1
Installed Roots	v2.2.10
Installed DataFrames	v1.8.0
Installed Pango_jll	v1.56.4+0
Installed SentinelArrays	v1.4.8
Installed StaticArraysCore	v1.4.3
Installed WeakRefStrings	v1.4.2
Installed WorkerUtilities	v1.6.1
Installed Ratios	v0.4.5
Installed DataStructures	v0.19.1
Installed ColorTypes	v0.11.5
Installed GraphRecipes	v0.5.14
Installed StatsBase	v0.34.6
Installed GeometryTypes	v0.8.5
Installed ColorSchemes	v3.31.0
Installed Graphs	v1.13.1
Installed Expat_jll	v2.7.1+0
Installed Inflate	v0.1.5
Installed Extents	v0.1.6
Installed TableTraits	v1.0.1
Installed Plots	v1.40.19
Installed PooledArrays	v1.4.3
Installed PrettyTables	v3.0.10
Installed ConstructionBase	v1.6.0
Installed ColorVectorSpace	v0.10.0
Installed DataValueInterfaces	v1.0.0
Installed AbstractTrees	v0.4.5
Installed ChainRulesCore	v1.26.0
Installed Ghostscript_jll	v9.55.1+0
Installed CompositionsBase	v0.1.2
Installed CSV	v0.10.15

Installed	Interpolations	v0.16.2
Installed	InverseFunctions	v0.1.17
Installed	CommonSolve	v0.2.4
Installed	Libuuid_jll	v2.41.1+0
Installed	SimpleTraits	v0.9.5
Installed	Crayons	v4.1.1
Installed	Adapt	v4.3.0
Installed	Latexify	v0.16.10
Installed	GeometryBasics	v0.5.10
Installed	OpenSSL_jll	v3.5.2+0
Installed	Tables	v1.12.1
Installed	IterTools	v1.10.0
Installed	IteratorInterfaceExtensions	v1.0.0
Installed	Glib_jll	v2.86.0+0
Installed	FilePathsBase	v0.9.24
Installed	AxisAlgorithms	v1.1.0
Installed	WoodburyMatrices	v1.0.0
Installed	SortingAlgorithms	v1.2.2
Installed	StringManipulation	v0.4.1
Precompiling project...		
1246.3 ms	InverseFunctions	
1432.3 ms	IteratorInterfaceExtensions	
1283.0 ms	WorkerUtilities	
1253.5 ms	CommonSolve	
1693.9 ms	OffsetArrays	
735.1 ms	DataValueInterfaces	
1191.8 ms	AbstractTrees	
1244.5 ms	IterTools	
880.6 ms	CompositionsBase	
1369.6 ms	InlineStrings	
969.1 ms	Extents	
891.6 ms	InvertedIndices	
905.0 ms	Ratios	
1084.8 ms	Inflate	
1057.4 ms	ConstructionBase	
1079.9 ms	Adapt	
928.8 ms	StaticArraysCore	
1040.7 ms	PooledArrays	
2503.5 ms	Crayons	
2443.8 ms	WoodburyMatrices	
3362.8 ms	DataStructures	
1660.0 ms	Preferences	
3006.6 ms	SentinelArrays	

1975.9 ms	SimpleTraits
4900.9 ms	ColorTypes
1503.7 ms	InverseFunctions → InverseFunctionsDatesExt
2888.8 ms	ChainRulesCore
2007.0 ms	InverseFunctions → InverseFunctionsTestExt
2519.0 ms	FilePathsBase
1060.0 ms	LogExpFunctions → LogExpFunctionsInverseFunctionsExt
821.6 ms	TableTraits
906.8 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
958.1 ms	Ratios → RatiosFixedPointNumbersExt
907.0 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
1667.5 ms	Unitful → InverseFunctionsUnitfulExt
1042.4 ms	OffsetArrays → OffsetArraysAdaptExt
1229.8 ms	Adapt → AdaptSparseArraysExt
1677.3 ms	Unitful → ConstructionBaseUnitfulExt
1214.8 ms	AxisAlgorithms
1100.3 ms	SortingAlgorithms
1157.7 ms	JLLWrappers
3174.1 ms	PrecompileTools
3336.0 ms	ChainRulesCore → ChainRulesCoreSparseArraysExt
3591.9 ms	ColorVectorSpace
1169.8 ms	FilePathsBase → FilePathsBaseMmapExt
1401.3 ms	Tables
4510.5 ms	LogExpFunctions → LogExpFunctionsChainRulesCoreExt
2368.7 ms	FilePathsBase → FilePathsBaseTestExt
1240.0 ms	OpenSSL_jll
1218.1 ms	Graphite2_jll
7688.0 ms	Colors
1155.8 ms	Libmount_jll
1203.8 ms	EpollShim_jll
4401.4 ms	Accessors
1434.2 ms	LLVMOpenMP_jll
4321.8 ms	StatsBase
1633.4 ms	Bzip2_jll
1228.6 ms	Xorg_libICE_jll
1169.6 ms	Xorg_libXau_jll
1259.0 ms	libpng_jll
1306.6 ms	libfdk_aac_jll
1394.6 ms	LAME_jll
1460.4 ms	LERC_jll
1402.6 ms	fzf_jll
1348.6 ms	EarCut_jll
1446.6 ms	JpegTurbo_jll

1065.6 ms	mtdev_jll
1440.7 ms	XZ_jll
1258.8 ms	Ogg_jll
1215.1 ms	Xorg_libXdmcp_jll
1538.6 ms	x265_jll
1475.5 ms	x264_jll
1565.9 ms	Zstd_jll
1580.2 ms	libaom_jll
1383.8 ms	Expat_jll
1597.8 ms	LZO_jll
1684.4 ms	Opus_jll
1354.0 ms	libevdev_jll
1370.3 ms	Xorg_xtrans_jll
1569.4 ms	Libiconv_jll
1365.8 ms	Libffi_jll
1255.9 ms	Libuuid_jll
1270.9 ms	eudev_jll
1427.7 ms	FriBidi_jll
3515.4 ms	RecipesBase
3271.3 ms	OpenSSL
6422.8 ms	StringManipulation
4059.6 ms	Accessors → LinearAlgebraExt
1438.3 ms	Accessors → TestExt
1705.1 ms	Accessors → UnitfulExt
1339.2 ms	Pixman_jll
6427.0 ms	ColorSchemes
1261.8 ms	Xorg_libSM_jll
1657.4 ms	FreeType2_jll
12469.3 ms	StaticArrays
2596.9 ms	JLFzf
1486.0 ms	libvorbis_jll
1013.2 ms	Xorg_libxcb_jll
2496.6 ms	Ghostscript_jll
1020.5 ms	DBus_jll
1646.6 ms	Libtiff_jll
979.6 ms	Wayland_jll
1583.7 ms	GettextRuntime_jll
942.1 ms	libinput_jll
7369.7 ms	Roots
21860.3 ms	Parsers
1931.2 ms	Fontconfig_jll
2444.3 ms	ArnoldiMethod
1720.3 ms	StaticArrays → StaticArraysChainRulesCoreExt

1514.2 ms	StaticArrays → StaticArraysStatisticsExt
1585.4 ms	ConstructionBase → ConstructionBaseStaticArraysExt
1558.9 ms	Adapt → AdaptStaticArraysExt
1870.9 ms	Accessors → StaticArraysExt
1339.9 ms	Xorg_xcb_util_jll
1589.7 ms	Xorg_libX11_jll
19468.9 ms	PlotUtils
2489.3 ms	Glib_jll
6085.3 ms	Latexify
1870.0 ms	Roots → RootsChainRulesCoreExt
1822.2 ms	Roots → RootsUnitfulExt
1426.9 ms	InlineStrings → ParsersExt
3122.1 ms	JSON
5235.3 ms	GeometryTypes
37082.2 ms	HTTP
10673.2 ms	Graphs
1252.0 ms	Xorg_xcb_util_image_jll
25567.2 ms	GeometryBasics
6424.1 ms	Interpolations
1205.3 ms	Xorg_xcb_util_keysyms_jll
1105.4 ms	Xorg_xcb_util_renderutil_jll
1252.7 ms	Xorg_xcb_util_wm_jll
1285.4 ms	Xorg_libXrender_jll
1382.0 ms	Xorg_libXext_jll
1369.6 ms	Xorg_libXfixes_jll
1659.7 ms	Xorg_libxkbfile_jll
2632.0 ms	Latexify → SparseArraysExt
4495.5 ms	UnitfulLatexify
2477.2 ms	WeakRefStrings
8585.5 ms	PlotThemes
1534.0 ms	Xorg_xcb_util_cursor_jll
4034.2 ms	Interpolations → InterpolationsUnitfulExt
5439.1 ms	NetworkLayout
2426.3 ms	Xorg_libXinerama_jll
2162.2 ms	Xorg_libXrandr_jll
2237.1 ms	Libglvnd_jll
12599.0 ms	RecipesPipeline
1490.1 ms	Xorg_libXcursor_jll
1675.8 ms	Xorg_libXi_jll
1572.2 ms	Xorg_xkbcomp_jll
2886.6 ms	Cairo_jll
1452.3 ms	Xorg_xkeyboard_config_jll
1305.3 ms	xkbcommon_jll

```

3194.9 ms HarfBuzz_jll
4491.8 ms NetworkLayout → NetworkLayoutGraphsExt
1449.5 ms Vulkan_Loader_jll
61243.7 ms PrettyTables
2423.5 ms libass_jll
2231.1 ms Pango_jll
3676.3 ms Qt6Base_jll
1193.9 ms libdecor_jll
4690.3 ms FFMPEG_jll
1320.6 ms GLFW_jll
2959.6 ms FFMPEG
6920.1 ms Qt6ShaderTools_jll
12957.1 ms GraphRecipes
2523.4 ms GR_jll
4714.6 ms Qt6Declarative_jll
1428.4 ms Qt6Wayland_jll
31857.1 ms CSV
8204.8 ms GR
64355.2 ms DataFrames
2560.0 ms Latexify → DataFramesExt
73897.5 ms Plots
4795.7 ms Plots → UnitfulExt
4929.8 ms Plots → GeometryBasicsExt
175 dependencies successfully precompiled in 222 seconds. 80 already precompiled.

```

```

Pkg.add("GraphRecipes")
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
using Random
using Plots
using GraphRecipes
using LaTeXStrings
using Distributions

```

```

Updating registry at `C:\Users\elrae\.julia\registries\General.toml`
Resolving package versions...
No Changes to `C:\Users\elrae\BEE4750\hw2-holly-1\Project.toml`
No Changes to `C:\Users\elrae\BEE4750\hw2-holly-1\Manifest.toml`

```

ArgumentError: Package Distributions not found in current path.
- Run `import Pkg; Pkg.add("Distributions")` to install the Distributions package.
ArgumentError: Package Distributions not found in current path.

- Run `import Pkg; Pkg.add("Distributions")` to install the Distributions package.

Stacktrace:

[1] macro expansion

@ .\loading.jl:2296 [inlined]

[2] macro expansion

@ .\lock.jl:273 [inlined]

[3] __require(into::Module, mod::Symbol)

@ Base .\loading.jl:2271

[4] #invoke_in_world#3

@ .\essentials.jl:1089 [inlined]

[5] invoke_in_world

@ .\essentials.jl:1086 [inlined]

[6] require(into::Module, mod::Symbol)

@ Base .\loading.jl:2260

[7] eval

@ .\boot.jl:430 [inlined]

[8] include_string(mapexpr::typeof(REPL.softscope), mod::Module, code::String, filename::String)

@ Base .\loading.jl:2734

[9] #invokelatest#2


```

@ .\essentials.jl:1055 [inlined]

[10] invokelatest

@ .\essentials.jl:1052 [inlined]

[11] (::VSCodeServer.var"#217#218"{VSCodeServer.NotebookRunCellArguments, String})()

@ VSCodeServer c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\run_cell.jl:1052 [inlined]

[12] withpath(f::VSCodeServer.var"#217#218"{VSCodeServer.NotebookRunCellArguments, String}, path::String)

@ VSCodeServer c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\run_cell.jl:1052 [inlined]

[13] notebook_runcell_request(conn::VSCodeServer.JSONRPC.JSONRPCEndpoint{Base.PipeEndpoint, Base.PipeEndpoint})

@ VSCodeServer c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\run_cell.jl:1052 [inlined]

[14] dispatch_msg(x::VSCodeServer.JSONRPC.JSONRPCEndpoint{Base.PipeEndpoint, Base.PipeEndpoint}, msg::String)

@ VSCodeServer.JSONRPC c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\run_cell.jl:1052 [inlined]

[15] serve_notebook(pipename::String, debugger_pipename::String, outputchannel_logger::Base.Logging.Logger)

@ VSCodeServer c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\run_cell.jl:1052 [inlined]

[16] top-level scope

@ c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\run_cell.jl:1052 [inlined]

```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

```

A = [0 0 1 ;
     1 0 1 ;

```

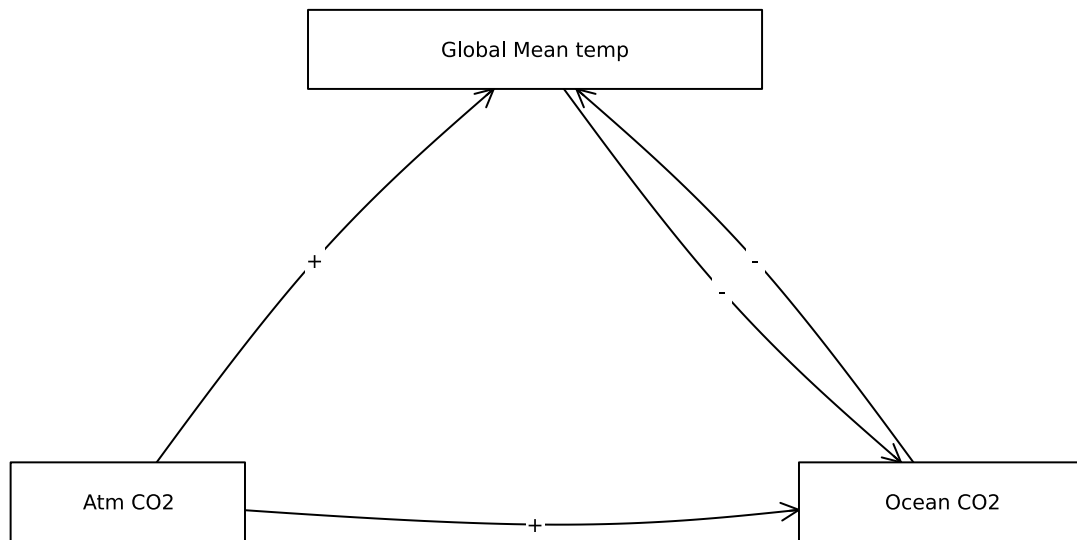
```

1 0 0 ]

names = ["Global Mean temp", "Atm CO2", "Ocean CO2"]
# modify this dictionary to add labels
edge_labels = Dict((2, 1) => "+", (1,3) => "-", (2,3) => "+", (3,1) => "-")
shapes=[:rect, :rect, :rect]
xpos = [0, -0.9, 0.9]
ypos = [1, 0, 0]

p = graphplot(A, names=names, edgelabel=edge_labels, markersize=0.15, markershapes=shapes, ma
display(p)

```



From Henry's law, we know that as the partial pressure of CO₂ increases above a body of water, the solubility of CO₂ in the water increases. This led us to choose a “+” for the relationship between Atm CO₂ and Ocean CO₂: As atm CO₂ increases, so will oceanic CO₂. On the other hand, as surface temperature increases, the ability of CO₂ to dissolve in water decreases. So as global mean temperature rises (along with ocean surface temperature), less CO₂ will dissolve, resulting in a “-” relationship. Additionally for this inverse, or dampening, relationship, oceanic carbon is a carbon sink, with more dissolved carbon resulting in slower

warming. This solidifies our selection of using “-”. We also know from the greenhouse effect that as more CO₂ resides in the atmosphere, more radiation will be trapped in the atmosphere contributing to a rise in temperature. Given this, we chose a “+” relationship.

Tip

Think about Henry’s law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of $2.3 \text{ kg}/(1000\text{m}^3)$. You can do this analytically or computationally.

Our system is not in compliance. After traveling 15 km to the second inflow point, the concentration is around $3.5 \text{ kg}/(1000\text{m}^3)$, which is above the regulatory limit. After mixing, it goes up even further. It is certainly not in compliance.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be $51 \text{ J/m}^2/\text{°C}$;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
# 3.2
# initial conditions/defining variables

A = 221.2
B = -1.3
C = 51
S = 1272

# albedo
ai = 0.5
a0 = 0.3

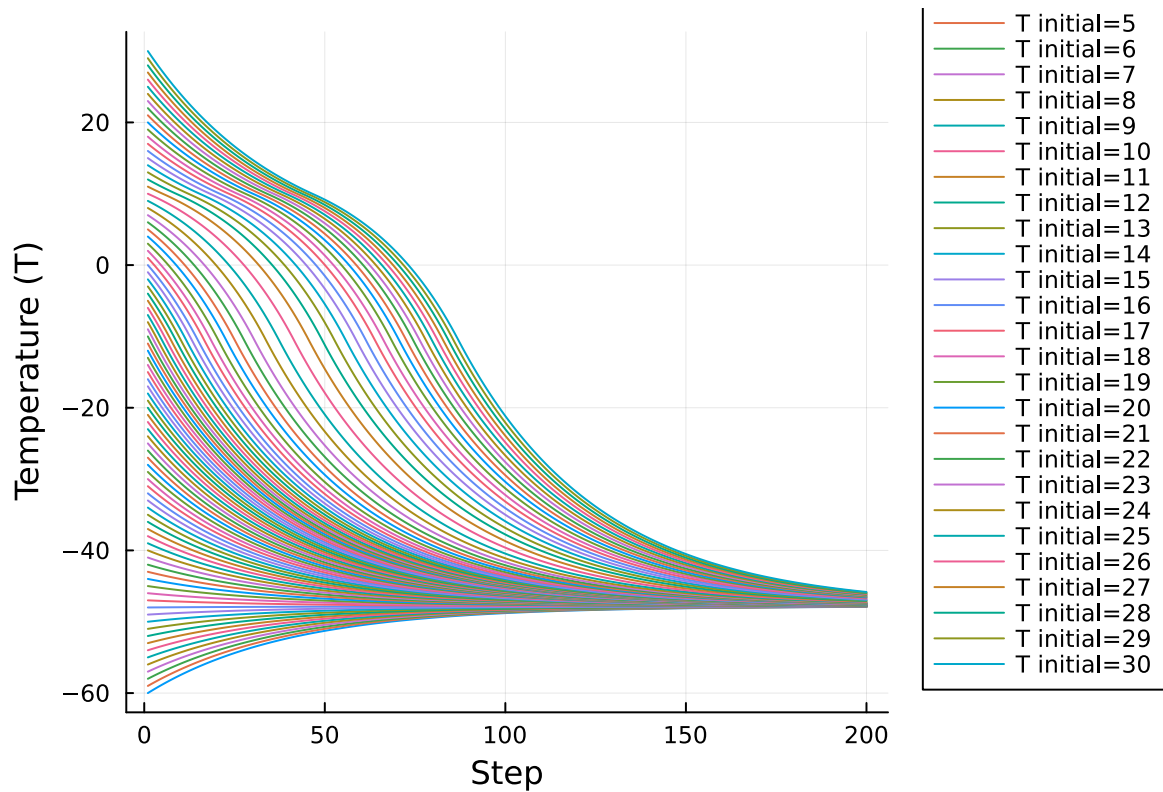
p = plot(xlabel="Step", ylabel="Temperature (T)", lw=2, legend = :outerbottomright)
# loop through T-values
for T0 in -60:1:30
    Ti = T0
    T = []
    I = []
    # 200 years of 1-year timesteps
    for i = 1:200
        push!(I, i)
        push!(T, Ti)
        # albedo based on temperature
        if Ti <= -10
            a = ai
        elseif Ti >= 10
            a = a0
        else
            a = ai + ((a0-ai)*((Ti+10)/20))
        end
        # new temperature
```

```

    T_new = Ti + 1/C * (((1-a)*S/4) - A + B*Ti)
    Ti = T_new

end
plot!(I, T, label="T initial=$T0")
end
display(p)

```



Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

Increasing the amount of solar radiation to $S = 1368$ creates another equilibrium point around 14 degrees C. This shows that an increase in solar radiation does lead to an increase in steady state temperature. It is possible that this increase in solar radiation could warm the planet to pre-industrial temperatures of 14 degrees C.

```

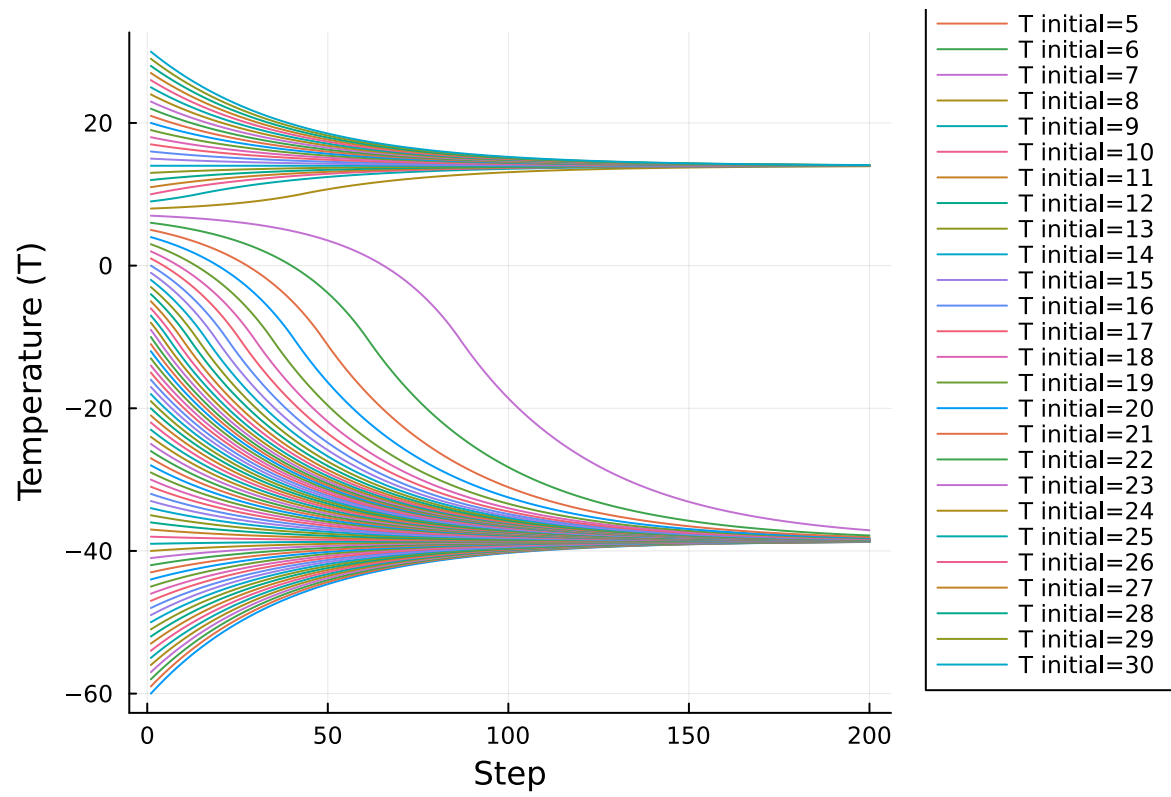
# 3.3
# initial conditions/defining variables

A = 221.2
B = -1.3
C = 51
S = 1368

# albedo
ai = 0.5
a0 = 0.3

p = plot(xlabel="Step", ylabel="Temperature (T)", lw=2, legend = :outerbottomright)
# loop through T-values
for T0 in -60:1:30
    Ti = T0
    T = []
    I = []
    # 200 years of 1-year timesteps
    for i = 1:200
        push!(I, i)
        push!(T, Ti)
        # albedo based on temperature
        if Ti <= -10
            a = ai
        elseif Ti >= 10
            a = a0
        else
            a = ai + ((a0-ai)*((Ti+10)/20))
        end
        # new temperature
        T_new = Ti + 1/C * (((1-a)*S/4) - A + B*Ti)
        Ti = T_new
    end
    plot!(I, T, label="T initial=$T0")
end
display(p)

```



References

List any external references consulted, including classmates.